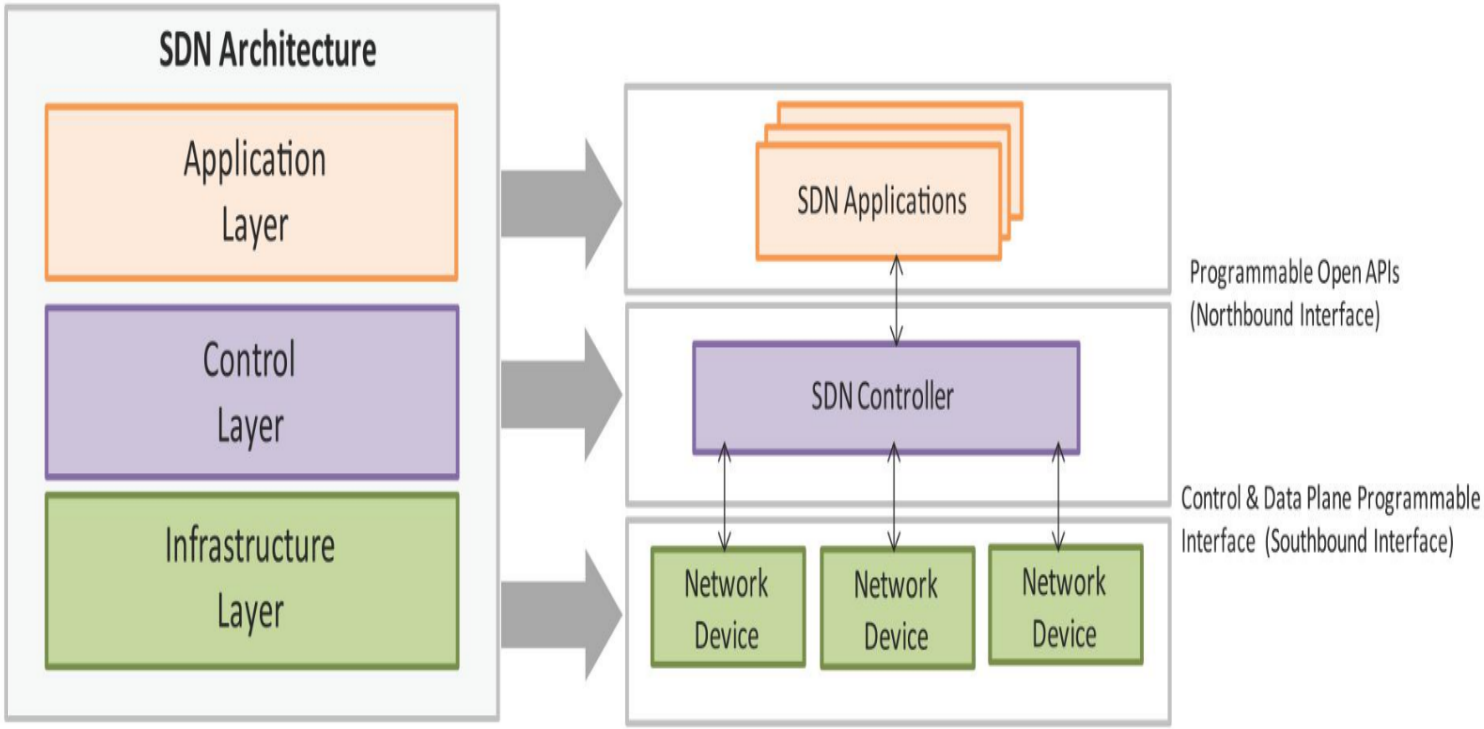
IT Workshop II (Cloud Computing): Cloud Concepts and Technologies, Python for Cloud

August 19, 2026

Software Defined Networking



SDN - Key Elements

- **Centralized Network Controller:** With decoupled the control and data planes and centralized network controller, the network administrators can rapidly configure the network.
- **Programmable Open APIs:** SDN architecture supports programmable open APIs for interface between the SDN application and control layers (Northbound interface). These open APIs that allow implementing various network services such as routing, quality of service (QoS), access control, etc.

SDN - Key Elements

- **Standard Communication Interface (OpenFlow):** SDN architecture uses a standard communication interface between the control and infrastructure layers (Southbound interface). OpenFlow, which is defined by the Open Networking Foundation (ONF) is the broadly accepted SDN protocol for the Southbound interface.

Python Program to Add Inbound Rule in Security Group

```
import boto3

ec2_client = boto3.client('ec2')

response = ec2_client.authorize_security_group_ingress(
    GroupId='enter security group id',
    IpPermissions=[
        {
            'IpProtocol':'tcp',
            'FromPort':22,
            'ToPort':22,
            'IpRanges':[{'CidrIp':'0.0.0.0/0'}]
        }
    ]
)

print(response)
```

Python Program to Describe EC2 Instance

```
import boto3
from pprint import pprint

ec2_client = boto3.client('ec2')

response = ec2_client.describe_instances(InstanceIds=['enter instance id'])

pprint(response)
```

Python Program for Amazon EC2 User Data

```
import boto3

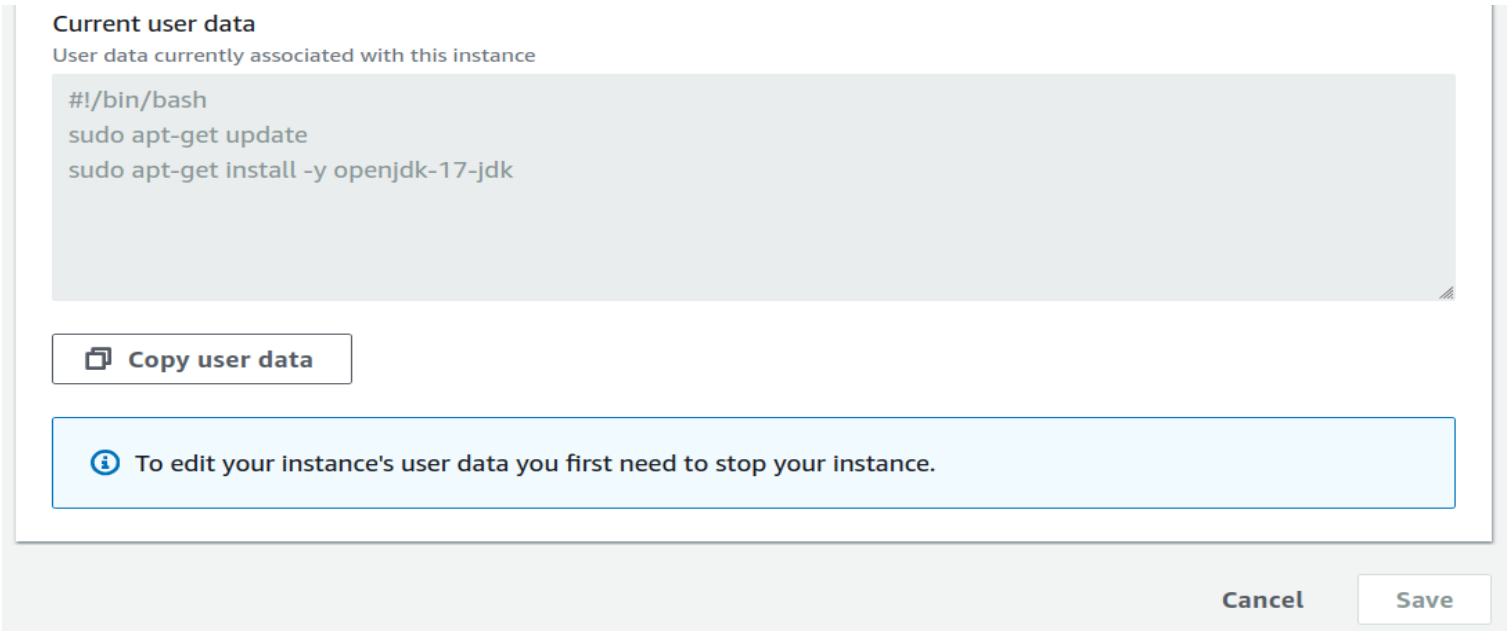
my_script = """#!/bin/bash
sudo apt-get update
sudo apt-get install -y openjdk-17-jdk
"""

ec2 = boto3.client("ec2",
                   "ap-south-1",
                    aws_access_key_id="enter access key id",
                    aws_secret_access_key="enter secret access key")

conn = ec2.run_instances(InstanceType="t2.micro",
                         MaxCount=1,
                         MinCount=1,
                         ImageId="ami-0522ab6e1ddcc7055",
                         KeyName="enter key name",
                         SecurityGroups=[
                          "enter security group name"
                         ],
                         UserData = my_script
                        )

print(conn)
```

Python Program for Amazon EC2 User Data



Python Program for Getting IP Address of Amazon EC2 Instance

```python
import boto3

def get_ip(instance_id):
    ec2_client = boto3.client('ec2')

    reservations = ec2_client.describe_instances(InstanceIds=[instance_id]).get('Reservations')

    for reservation in reservations:
        for instance in reservation['Instances']:
            print(instance.get('PublicIpAddress'))

get_ip('enter instance id')
```

Python Program to List Running EC2 Instances

```python
import boto3

ec2_resource = boto3.resource('ec2')
running_instances = ec2_resource.instances.filter(Filters=[{'Name': 'instance-state-name', 'Values':
['running']}])

for instance in running_instances:
    print(f"Instance ID: {instance.id}, State: {instance.state['Name']}")
```

OpenFlow

- OpenFlow is the broadly accepted SDN protocol for the Southbound interface.
- With OpenFlow, the forwarding plane of the network devices can be directly accessed and manipulated.
- OpenFlow uses the concept of flows to identify network traffic based on pre-defined match rules.
- Flows can be programmed statically or dynamically by the SDN control software.
- OpenFlow protocol is implemented on both sides of the interface between the controller and the network devices.

The End